

NAME

RDKitEnumerateCompoundLibrary.py - Enumerate a virtual compound library

SYNOPSIS

```
RDKitEnumerateCompoundLibrary.py [--colmode <collabel or colnum>] [--colRxnName <text or number>] [
--colRxnSMARTS <text or number>] [--compute2DCoords <yes or no>] [--infileParams <Name,Value,...>] [
--mode <RxnByName or RxnBySMARTS>] [--outfileParams <Name,Value,...>] [--overwrite] [
--prodMolNames <UseReactants or Sequential>] [--rxnName <text>] [--rxnNamesFile <FileName or auto>] [
--smartsRxn <text>] [--sanitize <yes or no>] [-w <dir>] -i <ReactantFile1,...> -o <outfile>
```

```
RDKitEnumerateCompoundLibrary.py [--colmode <collabel or colnum>] [--colRxnName <text or number>] [
--colRxnSMARTS <text or number>] [--rxnNamesFile <FileName or auto>] -l | --list
```

```
RDKitEnumerateCompoundLibrary.py -h | --help | -e | --examples
```

DESCRIPTION

Perform a combinatorial enumeration of a virtual library of molecules for a reaction specified using a reaction name or SMARTS pattern and reactant input files.

The SMARTS patterns for supported reactions names [Ref 134] are retrieved from file, ReactionNamesAndSMARTS.csv, available in MayaChemTools data directory. The current list of supported reaction names is shown below:

'1,2,4_triazole_acetohydrazide', '1,2,4_triazole_carboxylic_acid_ester', 3_nitrile_pyridine, Benzimidazole_derivatives_aldehyde, Benzimidazole_derivatives_carboxylic_acid_ester, Benzofuran, Benzothiazole, Benzothiophene, Benzoxazole_aromatic_aldehyde, Benzoxazole_carboxylic_acid, Buchwald_Hartwig, Decarboxylative_coupling, Fischer_indole, Friedlaender_chinoline, Grignard_alcohol, Grignard_carbonyl, Heck_non_terminal_vinyl, Heck_terminal_vinyl, Heteroaromatic_nuc_sub, Huisgen_Cu_catalyzed_1,4_subst, Huisgen_disubst_alkyne, Huisgen_Ru_catalyzed_1,5_subst, Imidazole, Indole, Mitsunobu_imide, Mitsunobu_phenole, Mitsunobu_sulfonamide, Mitsunobu_tetrazole_1, Mitsunobu_tetrazole_2, Mitsunobu_tetrazole_3, Mitsunobu_tetrazole_4, N_arylation_heterocycles, Negishi, Niemientowski_quinazoline, Nucl_sub_aromatic_ortho_nitro, Nucl_sub_aromatic_para_nitro, Oxadiazole, Paal_Knorr_pyrrole, Phthalazinone, Pictet_Spengler, Piperidine_indole, Pyrazole, Reductive_amination, Schotten_Baumann_amide, Sonogashira, Spiro_chromanone, Stille, Sulfon_amide, Suzuki, Tetrazole_connect_regioisomer_1, Tetrazole_connect_regioisomer_2, Tetrazole_terminal, Thiazole, Thiourea, Triaryl_imidazole, Urea, Williamson_ether, Wittig

The supported input file formats are: SD (.sdf, .sd), SMILES (.smi, .csv, .tsv, .txt)

The supported output file formats are: SD (.sdf, .sd), SMILES (.smi)

OPTIONS

-c, --colmode <collabel or colnum> [default: collabel]

Use column number or name for the specification of columns in a CSV file containing reaction names along with reaction SMARTS. You may specify a reaction names file using '--rxnNamesFile' option.

--colRxnName <text or number> [default: auto]

Column name or number corresponding to reaction names. The default value is automatically set based on the value of '-c, --colmode': 'RxnName' for 'collabel'; Reaction name column number for 'colnum'.

--colRxnSMARTS <text or number> [default: auto]

Column name or number corresponding to reaction SMARTS strings. The default value is automatically set based on the value of '-c, --colmode': 'RxnSMARTS' for 'collabel'; Reaction SMARTS column number for 'colnum'.

--compute2DCoords <yes or no> [default: yes]

Compute 2D coordinates of product molecules before writing them out.

-i, --infile <ReactantFile1, ReactantFile2...>

Comma delimited list of reactant file names for enumerating a compound library using reaction SMARTS. The number of reactant files must match number of reaction components in reaction SMARTS. All reactant input files must have the same format.

--infileParams <Name,Value,...> [default: auto]
A comma delimited list of parameter name and value pairs for reading molecules from files. The supported parameter names for different file formats, along with their default values, are shown below:
SD, MOL: removeHydrogens,yes,sanitize,yes,strictParsing,yes
SMILES: smilesColumn,1,smilesNameColumn,2,smilesDelimiter,space,smilesTitleLine,auto,sanitize,yes
Possible values for smilesDelimiter: space, comma or tab. These parameters apply to all reactant input files, which must have the same file format.

-e, --examples
Print examples.

-h, --help
Print this help message.

-l, --list
List available reaction names along with corresponding SMARTS patterns without performing any enumeration. In addition, reaction SMARTS patterns are validated.

-m, --mode <RxnByName or RxnBySMARTS> [default: RxnByName]
Indicate whether a reaction is specified by a reaction name or a SMARTS pattern. Possible values: RxnByName or RxnBySMARTS.

-o, --outfile <outfile>
Output file name.

--outfileParams <Name,Value,...> [default: auto]
A comma delimited list of parameter name and value pairs for writing molecules to files. The supported parameter names for different file formats, along with their default values, are shown below:
SD: kekulize,yes,forceV3000,no
SMILES: smilesKekulize,no,smilesDelimiter,space, smilesIsomeric,yes,smilesTitleLine,yes

-p, --prodMolNames <UseReactants or Sequential> [default: UseReactants]
Generate names of product molecules using reactant names or assign names in a sequential order. Possible values: UseReactants or Sequential. Format of molecule names: UseReactants - <ReactName1>_<ReactName2>..._Prod<Num>; Sequential - Prod<Num>

--overwrite
Overwrite existing files.

-r, --rxnName <text>
Name of a reaction to use for enumerating a compound library. This option is only used during 'RxnByName' value of '-m, --mode' option.

--rxnNamesFile <FileName or auto> [default: auto]
Specify a file name containing data for names of reactions and SMARTS patterns or use default file, ReactionNamesAndSMARTS.csv, available in MayaChemTools data directory.
Default reactions SMARTS file format: RxnName,RxnSMARTS.
The local file format is assumed to be same as the default file format. You may explicitly specify column names or numbers for reaction name and reaction SMARTS using '--colRxnName' and '--colRxnSMARTS' options.

-s, --smartsRxn <text>
SMARTS pattern of a reaction to use for enumerating a compound library. This option is only used during 'RxnBySMARTS' value of '-m, --mode' option.

--sanitize <yes or no> [default: yes]

Sanitize product molecules before writing them out.

-w, --workingdir <dir>

Location of working directory which defaults to the current directory.

EXAMPLES

To list all available reaction names along with their SMARTS pattern, type:

```
% RDKitEnumerateCompoundLibrary.py -l
```

To perform a combinatorial enumeration of a virtual compound library corresponding to named amide reaction, Schotten_Baumann_amide, and write out a SMILES file type:

```
% RDKitEnumerateCompoundLibrary.py -r Schotten_Baumann_amide  
-i 'SampleAcids.smi,SampleAmines.smi' -o SampleOutCmpdLibrary.smi
```

To run the previous command using a local reaction names file with explicit specification of column names containing reaction names and SMARTS, and write out a SMILES file type:

```
% RDKitEnumerateCompoundLibrary.py -r Schotten_Baumann_amide  
--rxnNamesFile ReactionNamesAndSMARTS.csv  
--colmode collabel --colRxnName RxnName --colRxnSMARTS RxnSMARTS  
-i 'SampleAcids.smi,SampleAmines.smi' -o SampleOutCmpdLibrary.smi
```

To perform a combinatorial enumeration of a virtual compound library corresponding to an amide reaction specified using a SMARTS pattern and write out a SD file containing sanitized molecules, computed 2D coordinates, and generation of molecule names from reactant names, type:

```
% RDKitEnumerateCompoundLibrary.py -m RxnBySMARTS  
-s '[O:2]=[C:1][OH].[N:3]>>[O:2]=[C:1][N:3]'  
-i 'SampleAcids.smi,SampleAmines.smi' -o SampleOutCmpdLibrary.sdf
```

To perform a combinatorial enumeration of a virtual compound library corresponding to an amide reaction specified using a SMARTS pattern and write out a SD file containing unsanitized molecules, without generating 2D coordinates, and a sequential generation of molecule names, type:

```
% RDKitEnumerateCompoundLibrary.py -m RxnBySMARTS -c no --sanitize no  
-p Sequential -s '[O:2]=[C:1][OH].[N:3]>>[O:2]=[C:1][N:3]'  
-i 'SampleAcids.smi,SampleAmines.smi' -o SampleOutCmpdLibrary.sdf
```

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SEE ALSO

RDKitConvertFileFormat.py, RDKitFilterPAINS.py, RDKitSearchFunctionalGroups.py, RDKitSearchSMARTS.py

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The functionality available in this script is implemented using RDKit, an open source toolkit for cheminformatics developed by Greg Landrum.

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